RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Course Code	IS353IA	Maximum Marks	10+50
Sem	V	CIE 2	
UG/PG	UG	Faculty: AS/VH/VG/JS/SHRS/ARA/MEM	

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
(Common to CSE/ISE/CD/CY)

SCHEME AND SOLUTION

Q. No.	QUIZ	M		
1	Abstraction	1		
2	d times	1		
3	Effective branching factor	2		
4	Weight the influence of each nearest neighbor xi according to its distance: wi = 1/d(x', xi)$^{2.}$	1		
5	If Xi is an irrelevant attribute, then P(Xi|Y) becomes almost uniformly distributed for every class y. The class-conditional probabilities for every class thus receive similar contributions of P(Xi|Y), resulting in negligible impact on the posterior probability estimates.	2		
6	Completeness, Cost Optimality, Time complexity, Space Complexity (0.5x4 = 2 Marks)	2		
7	STATE: the state in the state space to which the node corresponds; • PARENT: the node in the search tree that generated this node; •ACTION: the action that was applied to the parent to generate the node; •PATH-COST: the cost, traditionally denoted by g(n), of the path from the initial state to the node, as indicated by the parent pointers. (0.5x4 = 2 Marks)	2		

Q. No.	TEST	M
1	Apply BFS and DFS to the following graph and compare the results . Show all the necessary steps during the traversal and reasons **Answer** BFS : {A B S C G D E F H} (3*2 = 6 Marks) DFS: {A B S C D E H G F} **Advantages of BFS** • **Optimal Solution** • **Completeness –** it will find the solution if it exists. • **Uniform cost search –** BFS can be modified to search for the shortest path in a weighted graph by replacing the queue data structure with a priority queue. • **Level-wise traversal –** BFS visits all the nodes at a given level before moving to the next level. This approach makes it useful in certain scenarios such as finding the shortest path or exploring a game state. Disadvantages of BFS • Space complexity • Time complexity – BFS can be slow for graphs with a large number of nodes or edges. .Advantages of DFS • **Memory-efficient –** DFS uses less memory than BFS. • **Time Efficient –** For graphs with many nodes or edges, DFS may be faster than BFS. • **Depth-first search –** useful in certain scenarios, such as finding the maximum depth of a tree or graph. **Disadvantages of DFS** • **Completeness –** DFS is not guaranteed to find the solution if it exists. • **Non-optimal solution –** DFS may not find the shortest path between two nodes in an unweighted graph. • **Local minimum –** DFS may get stuck in a local minimum in a weighted graph.	

2	A* Algorithm calculates f(E), f(H) and f(J). f(E) = (3+1+3+5) + 3 = 15 f(H) = (3+1+3+2) + 3 = 12 f(J) = (3+1+3+3) + 0 = 10 Since f(J) is least, so it decides to go to node J. Path- A → F → G → I → J This is the required shortest path from node A to node J.	10
3	**a**. *False*: a lucky DFS might expand exactly d nodes to reach the goal. A∗ largely dominates any graph-search algorithm that is *guaranteed to find optimal solutions*. **b**. *True*: h(n) = 0 is always an admissible heuristic, since costs are nonnegative. **c**. *True*: A* search is often used in robotics; the space can be discretized or skeletonized. **d**. *True*: depth of the solution matters for breadth-first search, not cost. **e**. *False*: a rook canmove across the board in move one, although the Manhattan distance from start to finish is 8.	10 (2x5)
4	$P(A=1\|-)=2/5=0.4,\ P(B=1\|-)=2/5=0.4,$ $P(C=1\|-)=1,\ P(A=0\|-)=3/5=0.6,$ $P(B=0\|-)=3/5=0.6, P(C=0\|-)=0;\ P(A=1\|+)=3/5=0.6,$ $P(B=1\|+)=1/5=0.2,\ P(C=1\|+)=2/5=0.4,$ $P(A=0\|+)=2/5=0.4,\ P(B=0\|+)=4/5=0.8,$ $P(C=0\|+)=3/5=0.6.$ b) Let $P(A=0,B=1,C=0)=K$. $\begin{aligned} & P(+\|A=0,B=1,C=0) \\ = & \frac{P(A=0,B=1,C=0\|+)\times P(+)}{P(A=0,B=1,C=0)} \\ = & \frac{P(A=0\|+)P(B=1\|+)P(C=0\|+)\times P(+)}{K} \\ = & 0.4\times 0.2\times 0.6\times 0.5/K \\ = & 0.024/K. \end{aligned}$ $\begin{aligned} & P(-\|A=0,B=1,C=0) \\ = & \frac{P(A=0,B=1,C=0\|-)\times P(-)}{P(A=0,B=1,C=0)} \\ = & \frac{P(A=0\|-)\times P(B=1\|-)\times P(C=0\|-)\times P(-)}{K} \\ = & 0/K \end{aligned}$ The class label should be '+'.	10 2.5*4 Marks

C

$$P(A=0|+) = (2+2)/(5+4) = 4/9,$$
$$P(A=0|-) = (3+2)/(5+4) = 5/9,$$
$$P(B=1|+) = (1+2)/(5+4) = 3/9,$$
$$P(B=1|-) = (2+2)/(5+4) = 4/9,$$
$$P(C=0|+) = (3+2)/(5+4) = 5/9,$$
$$P(C=0|-) = (0+2)/(5+4) = 2/9.$$

D

$$\begin{aligned} & P(+|A=0,B=1,C=0) \\ = & \frac{P(A=0,B=1,C=0|+) \times P(+)}{P(A=0,B=1,C=0)} \\ = & \frac{P(A=0|+)P(B=1|+)P(C=0|+) \times P(+)}{K} \\ = & \frac{(4/9) \times (3/9) \times (5/9) \times 0.5}{K} \\ = & 0.0412/K \end{aligned}$$

$$\begin{aligned} & P(-|A=0,B=1,C=0) \\ = & \frac{P(A=0,B=1,C=0|-) \times P(-)}{P(A=0,B=1,C=0)} \\ = & \frac{P(A=0|-) \times P(B=1|-) \times P(C=0|-) \times P(-)}{K} \\ = & \frac{(5/9) \times (4/9) \times (2/9) \times 0.5}{K} \\ = & 0.0274/K \end{aligned}$$

The class label should be '+'.

5

$\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$

Brightness	Saturation	Class	Distance
10	25	Red	10
40	20	Red	25
50	50	Blue	33.54
25	80	Blue	45
60	10	Red	47.17
70	70	Blue	61.03
60	90	Blue	68.01

Class = Red

5

	• Nearest neighbor classification is part of a more general technique known as instance-based learning, which does not build a global model, but rather uses the training examples to make predictions for a test instance. • Although lazy learners, such as nearest neighbor classifiers, do not require model building, classifying a test instance can be quite expensive • Nearest neighbor classifiers make their predictions based on local information. • Nearest neighbor classifiers can produce decision boundaries of arbitrary Shape • Nearest neighbor classifiers have difficulty handling missing values in both the training and test sets since proximity computations normally require the presence of all attributes • Nearest neighbor classifiers can handle the presence of interacting attributes Any 5 (1*5)	